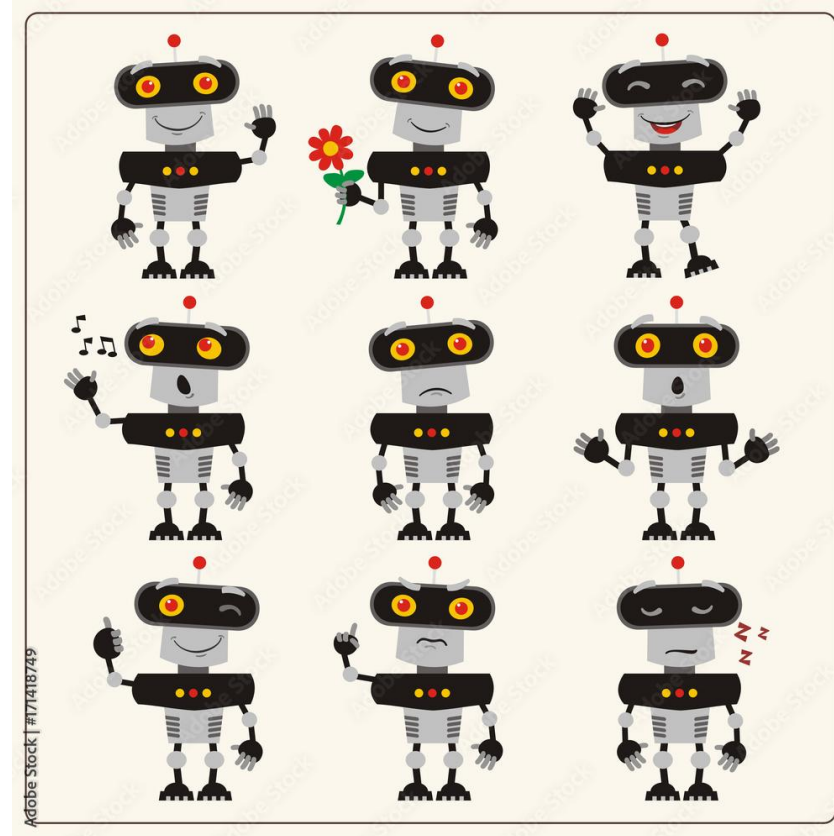


Males and Females (Machine) Learning to Express Emotions More Granularly

Bradley Allen



<https://stock.adobe.com/images/set-emoticon-black-gray-robot-with-different-emotions-collection-isolated-robots-in-various-poses-in-cartoon-style/171418749>

$$\frac{TP}{TP + FN}$$

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*When we can talk about
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Emotional Granularity

- Improves regulation of distress and other negative emotions
- Associated with healthier coping skills
- More frequent regulation, increased resilience

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Problem

- Estimated 36% of people accurately identify their emotions
- Polling from 2024 to 2026 consistently shows heightened depression, anxiety, and stress

A man with grey hair, wearing a red cardigan over a blue shirt and a patterned tie, is sitting on a wooden chair. He is smiling and looking towards the camera. A speech bubble is positioned above him, containing text. In the background, there is a potted plant and a wall with some artwork.

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- Estimated 36% of people accurately identify their emotions
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A photograph of a middle-aged man with grey hair, smiling and sitting on a wooden chair. He is wearing a bright red zip-up cardigan over a light blue shirt and a dark patterned tie. He is also wearing blue jeans and white sneakers. A speech bubble is positioned above him, containing text. In the background, there is a large green plant and a purple wall.

When we can talk about our feelings, they become less overwhelming, less upsetting, and less scary.

Research Questions

Do machine models support discriminant validity of DASS-21 constructs via important feature extraction?

Do important features vary between males and females with respect to DASS-21 subscales and demographic features as predictors of depression, anxiety, and stress?

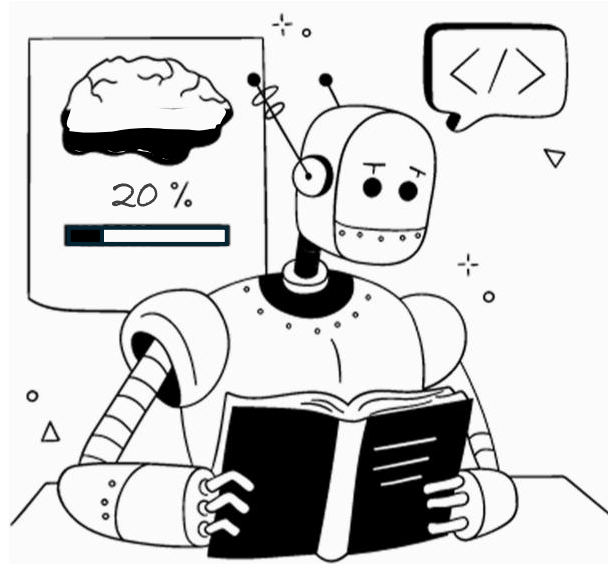
If important features vary by gender, what might this suggest about learning a clinically aligned vocabulary?

Item no.	SCALE Subscale Item	Root Vocabulary	Item no.	SCALE Subscale Item	Root Vocabulary
	DEPRESSION			STRESS	
	<i>Anhedonia</i>	Anhedonia		<i>Difficulty relaxing</i>	Stress
3	I couldn't seem to experience any positive feeling at all	Positive feeling	8	I found it difficult to relax	Relaxing
	<i>Devaluation of life</i>	Devalue	22	I found it hard to wind down	Wind-down
38	I felt that life was meaningless	Meaningless		<i>Easily upset / agitated</i>	Upset, agitated
	<i>Dysphoria</i>	Dysphoria	11	I found myself getting agitated	
26	I felt down-hearted and blue	Down-hearted, blue		<i>Impatient</i>	Impatient
	<i>Hopelessness</i>	Hopeless	35	I was intolerant of anything that kept me from getting on with what I was doing	Intolerant
10	I felt that I had nothing to look forward to	Look forward to		<i>Irritable / over-reactive</i>	Irritable, over-react
	<i>Inertia</i>	Inertia	6	I tended to over-react to situations	
42	I found it difficult to work up the initiative to do things	Initiative	18	I felt I was rather touchy	Touchy
	<i>Lack of interest / involvement</i>	Interest, involvement		<i>Nervous arousal</i>	
31	I was unable to become enthusiastic about anything	Enthusiasm	12	I felt that I was using a lot of nervous energy	Nervous
	<i>Self-deprecation</i>	Self-deprecation			
17	I felt that I wasn't worth much as a person	Worth			
	ANXIETY	Anxiety			
	<i>Autonomic arousal</i>				
2	I was aware of dryness of my mouth	Dry mouth			
4	I experienced breathing difficulty (e.g., excessively rapid breathing, breathlessness in the absence of physical exertion)	Difficulty breathing			
25	I was aware of the action of my heart in the absence of physical exertion (e.g., sense of heart rate increase, heart missing a beat)	Heartbeat			
	<i>Situational anxiety</i>				
40	I was worried about situations in which I might panic and make a fool of myself	Panic, fool			
	<i>Skeletal musculature effects</i>				
41	I experienced trembling (e.g., in the hands)	Trembling			
	<i>Subjective experience of anxious affect</i>				
20	I felt scared without any good reason	Scared			
28	I felt I was close to panic	Panic			

Note: Item numbers are based on the DASS-42 survey items included in the Answers to the Depression Anxiety Stress Scales data set. The items in this table are those related to the DASS-21 only.

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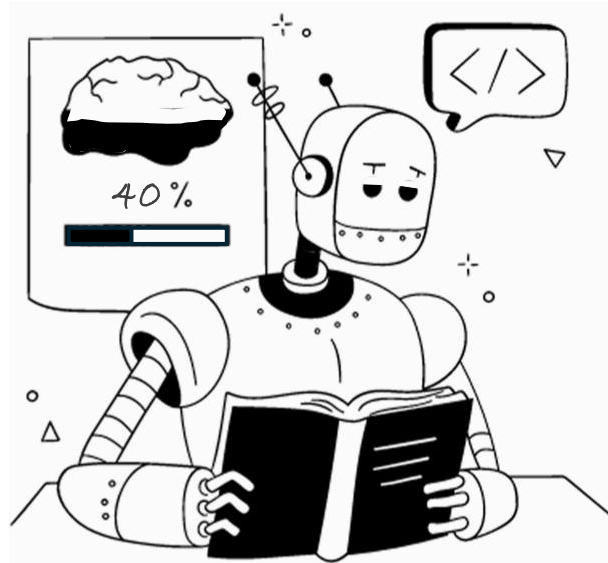
Machine Learning



Meshine Lerning

DATA

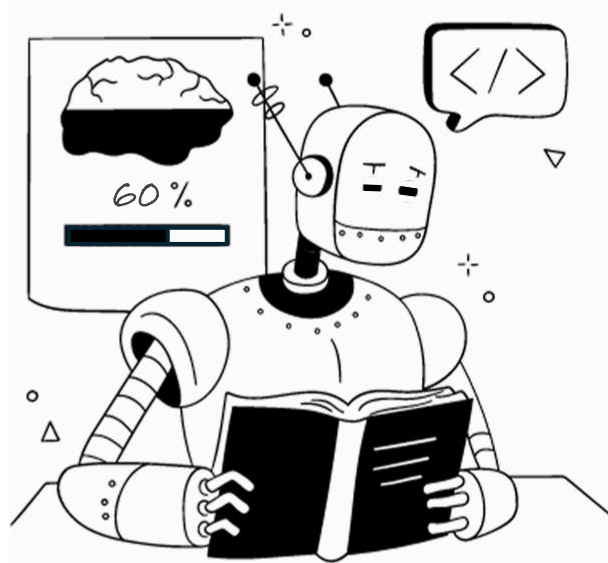
- Open Psychometrics Answers to the Depression Anxiety Stress Scales data
- 39,775 responses
- 172 variables
 - Depression Anxiety Stress Scales
 - Ten-Item Personality Inventory
 - Demographic variables (e.g., gender, race, age, etc.)



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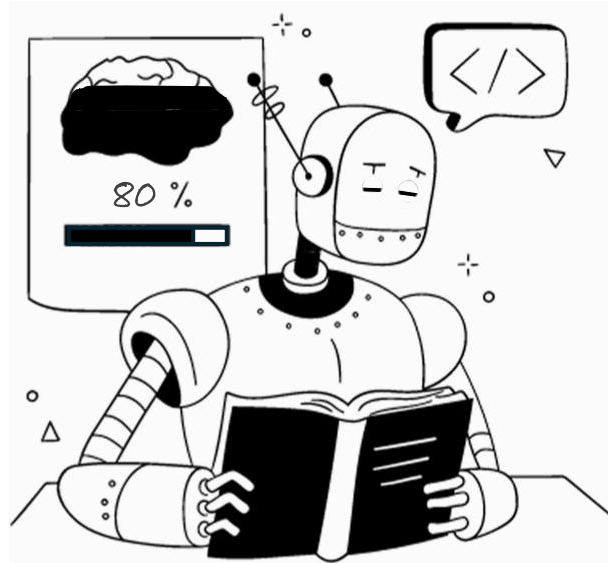
MODELS

- Seven models:
 - Random Forest
 - AdaBoost
 - Gradient Boosting
 - Histogram Gradient Boosting
 - XGBoost
 - LightGBM
 - CatBoost
- Apply Ensemble

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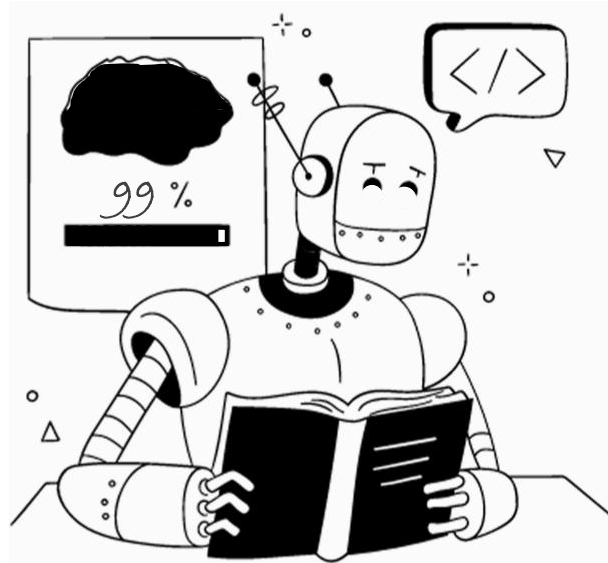
CLASSIFICATION MODELING

- Clinical cutoff scores
 - Negative (0 = normal to mild symptoms)
 - Positive (1 = moderate to severe symptoms)

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EXTRACT IMPORTANT FEATURES

- Subscales of DASS-21 predicting depression, anxiety, stress
- Demographic variables predicting depression, anxiety, stress





Cost Sensitive Machine Models

Minimize misclassifications (FP, FN)

(Feng et al., 2026; Sterner et al., 2025)

- Grid Search CV
 - Tree depth
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Improve model efficiency and bias correction

(Athey et al., 2019)

- Minimum sample splits ($n = 30$)
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Replicability

- Models were trained on 80% of the data, and test on 20%.
- Where possible, random state = 42 was applied
- Results and Jupyter Notebook included



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Ensemble

- Best model parameters passed in
- Soft voting to capture nuance of human behavior (Gupta & Pandit, 2026; Jain et al., 2025)



```
"Random Forest": (  
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),  
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),  
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        "max_features": ["sqrt", 0.1, 0.3, 0.5, None],  
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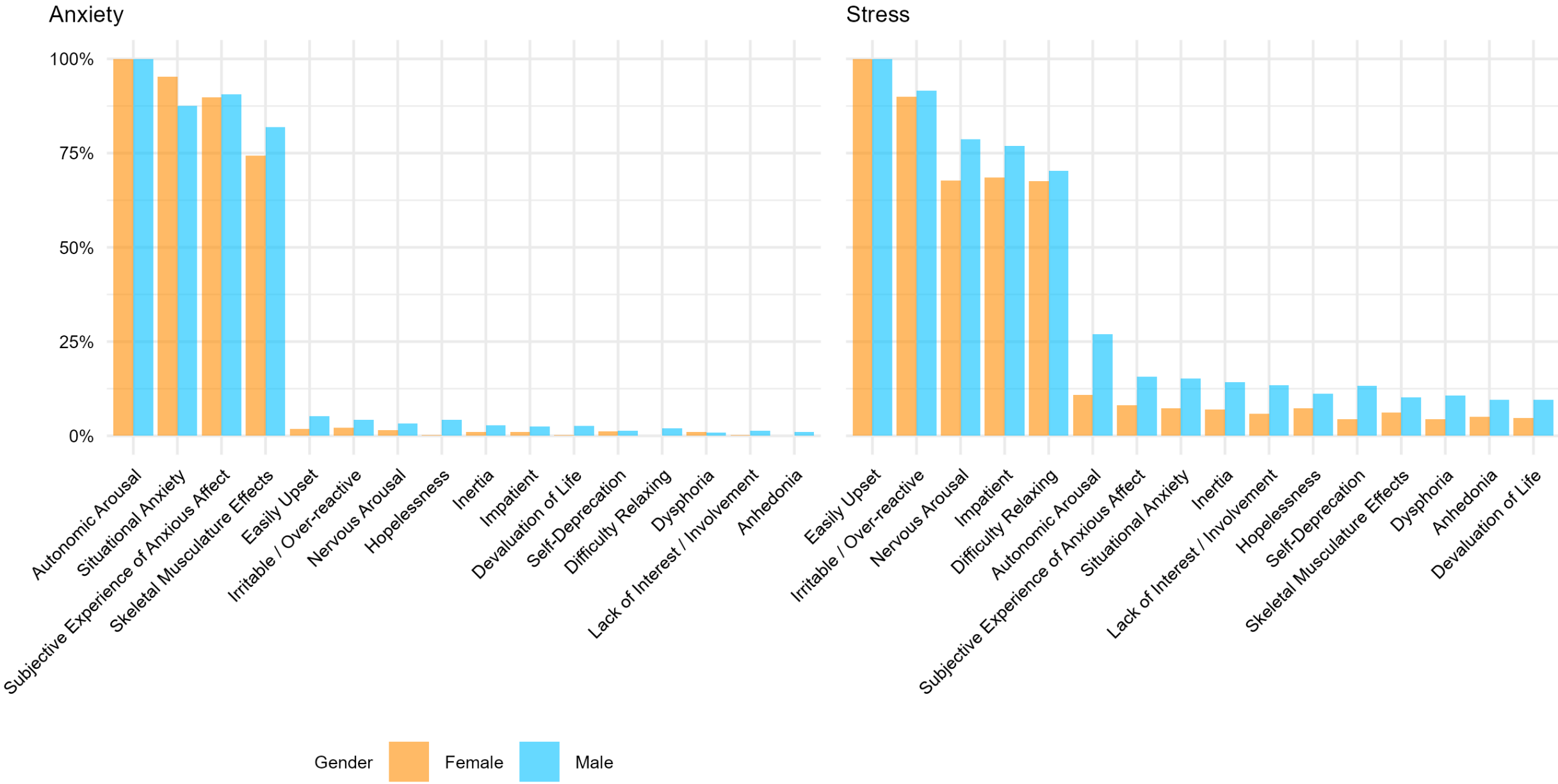
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        "colsample_bytree": [0.3, 0.5, 1.0],  
        "class_weight": ["balanced"],  
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Results

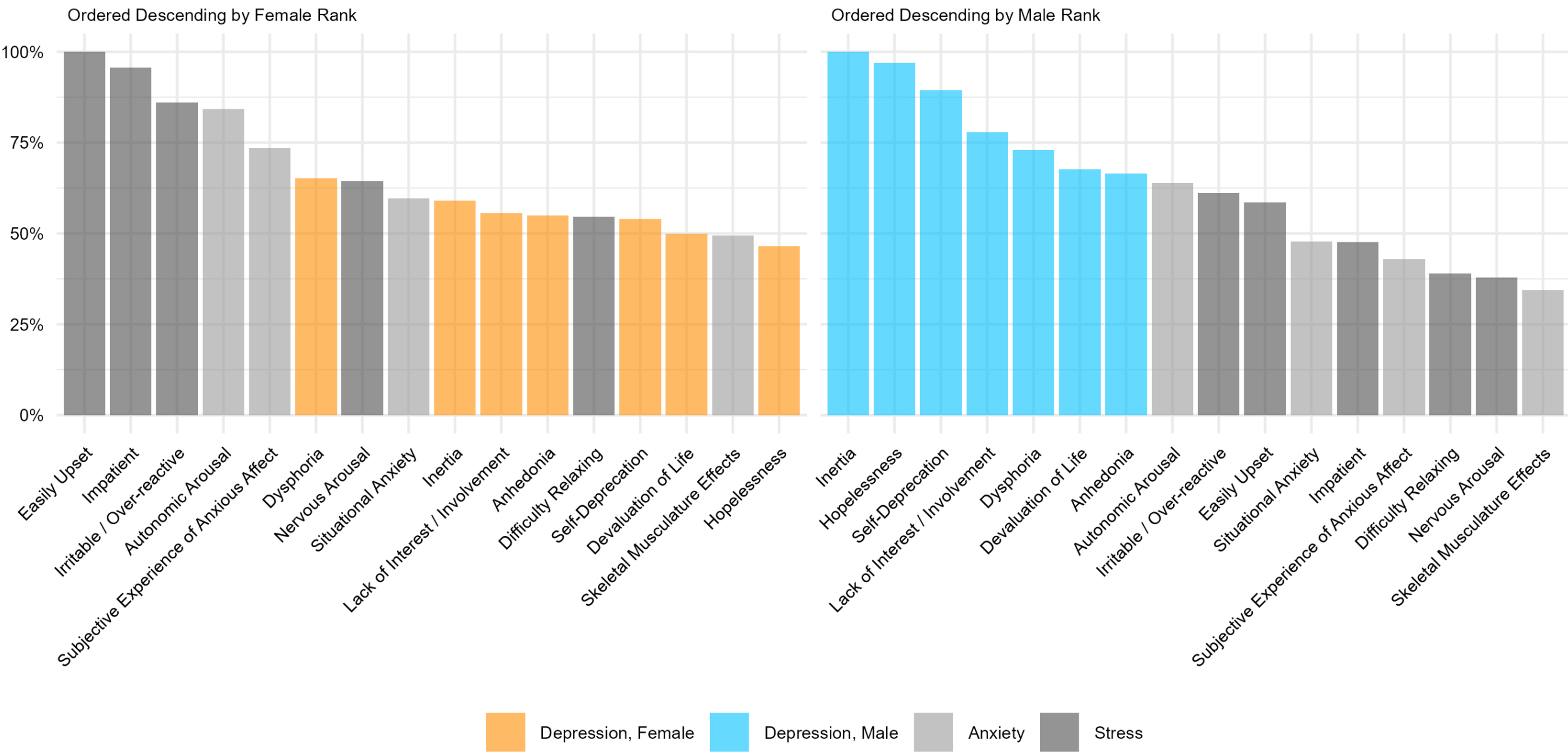
Table 5				
Ensemble Model Results for <u>DASS-21 Subscales</u> as Predictors of Depression, Anxiety, and Stress				
	Accuracy	Precision	Recall	F1 Score
Depression				
Female	1	1	1	1
Male	.99	.99	.99	.99
Anxiety				
Female	1	1	1	1
Male	1	1	1	1
Stress				
Female	1	1	1	1
Male	1	1	1	1
Note: All reported scores are based on weighted averages				

Table 7				
Ensemble Model Results for <u>Demographic Variables</u> as Predictors of Depression, Anxiety, and Stress				
	Accuracy	Precision	Recall	F1 Score
Depression				
Female	.76	.76	.76	.76
Male	.76	.76	.76	.76
Anxiety				
Female	.74	.73	.74	.74
Male	.72	.72	.72	.72
Stress				
Female	.75	.75	.75	.75
Male	.74	.74	.74	.74
Note: All reported scores are based on weighted averages				

DASS-21



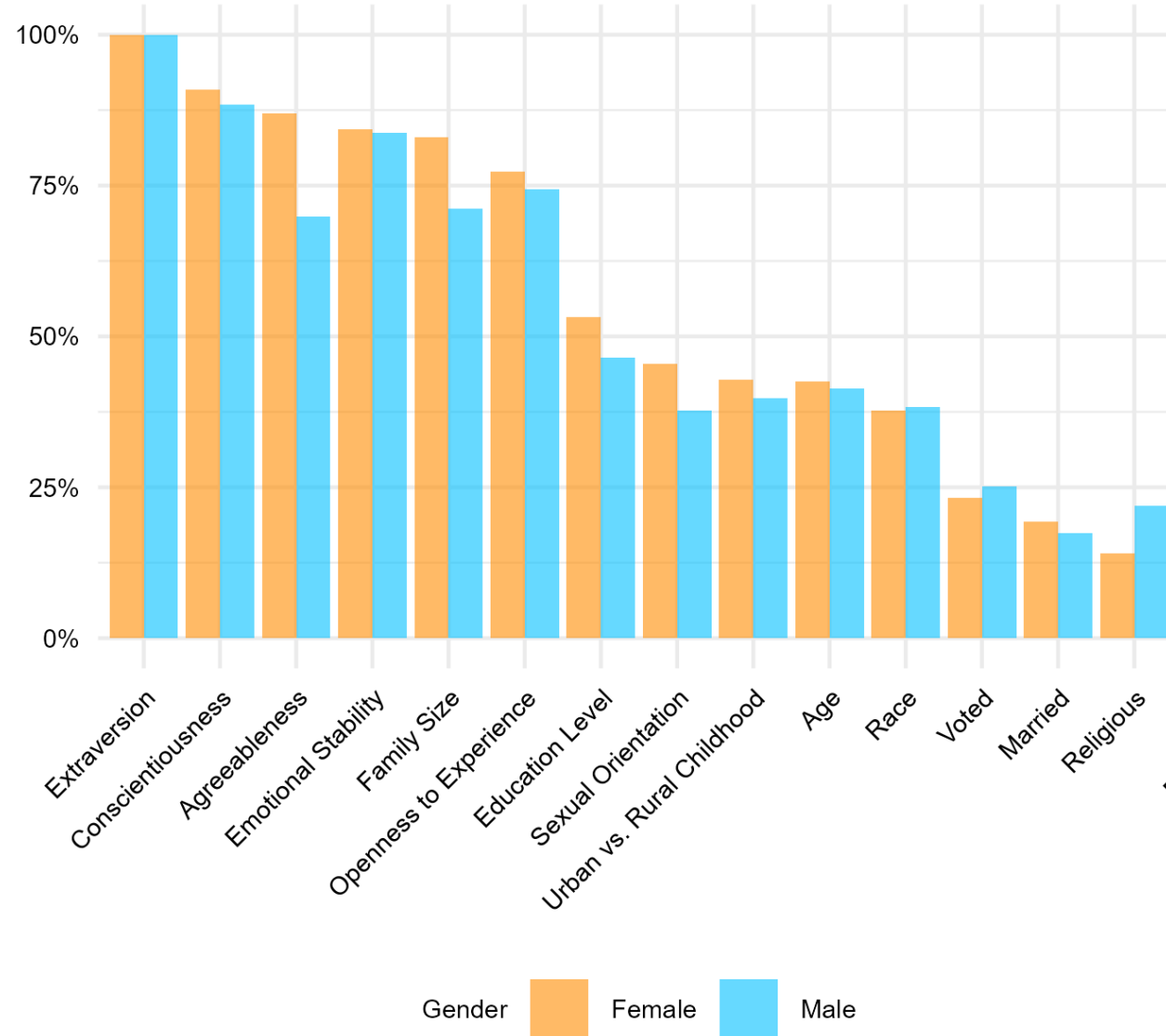
DASS-21



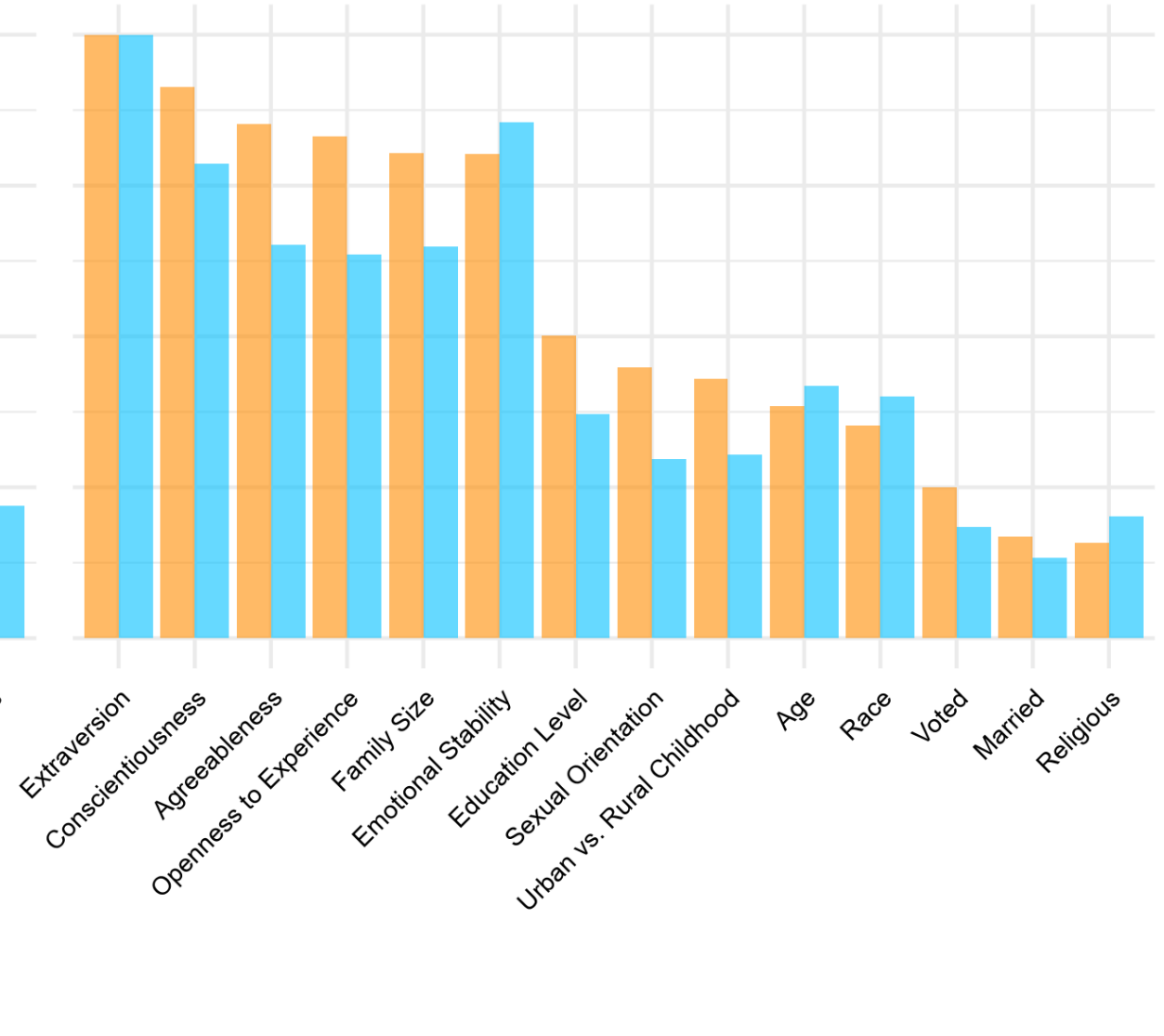
Demographic

Depression

Ordered Descending by Female Rank



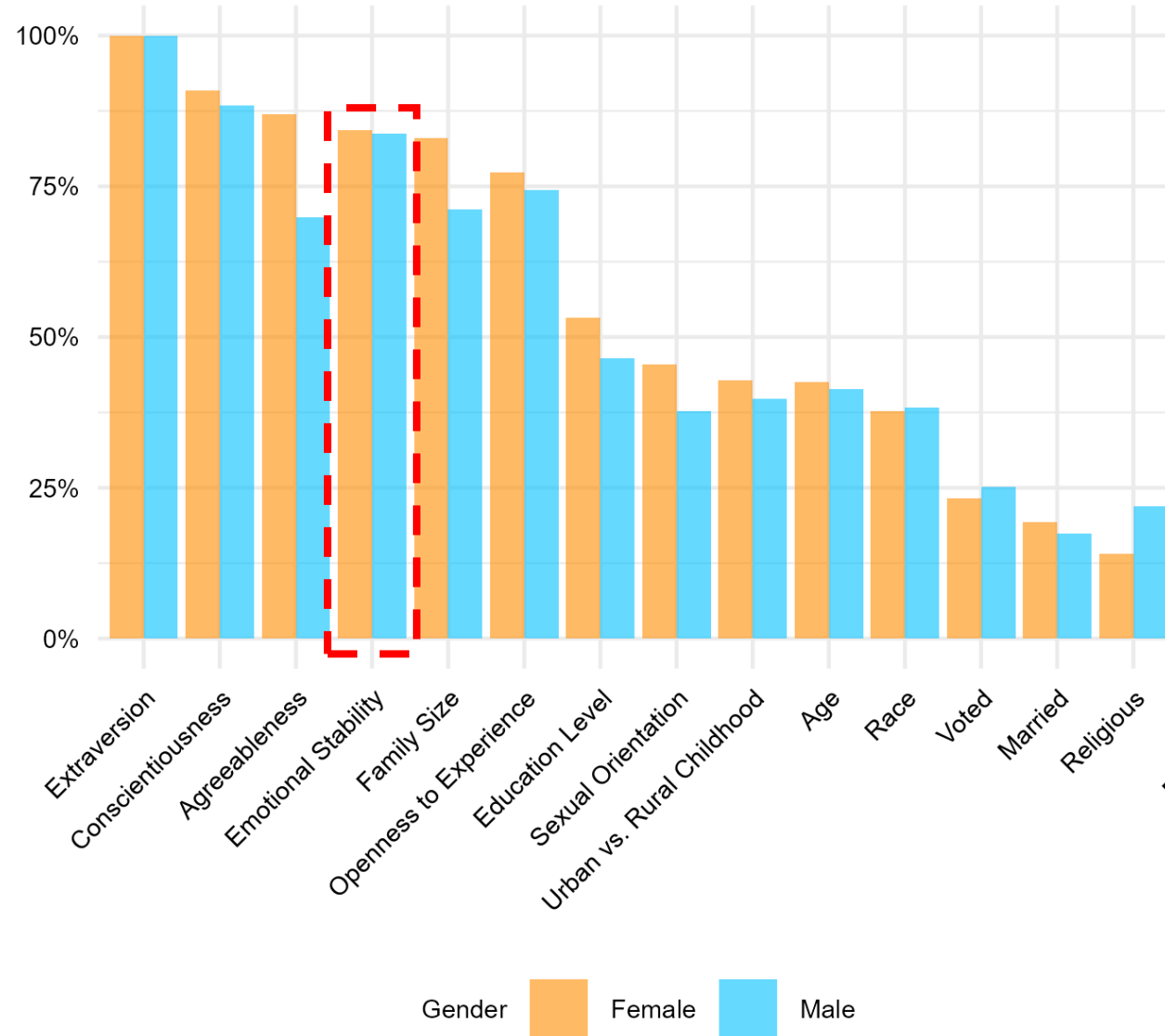
Anxiety



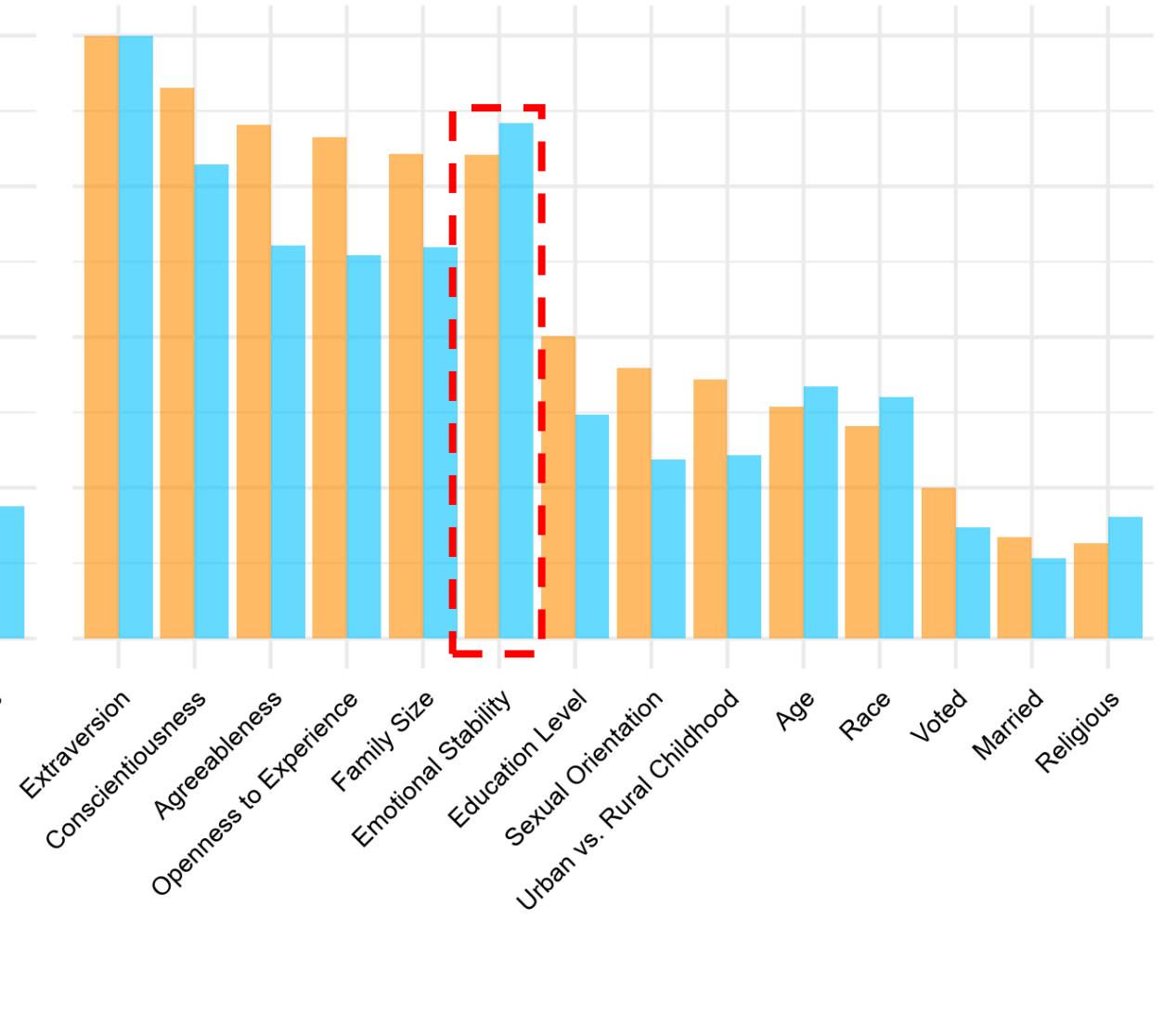
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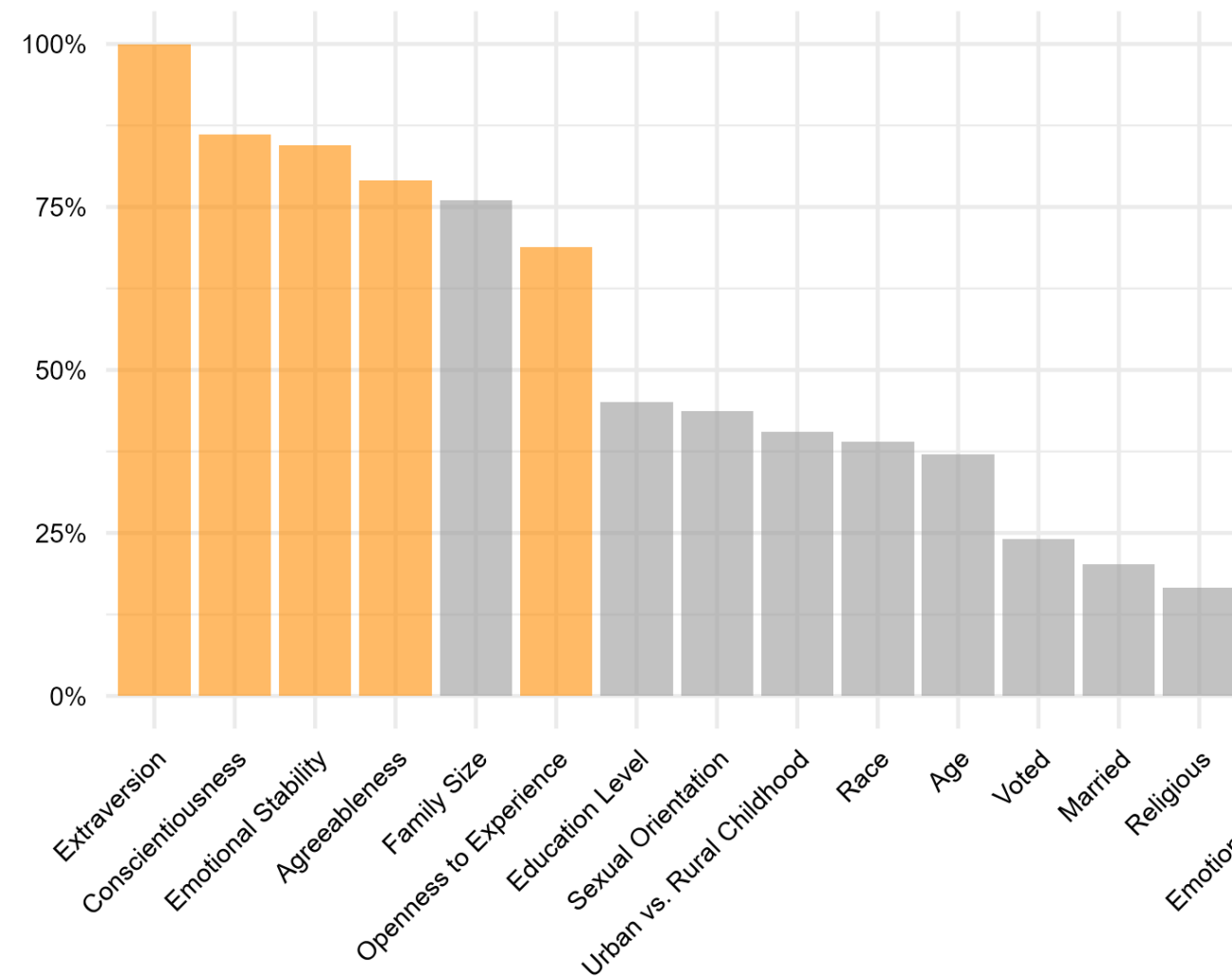


Anxiety

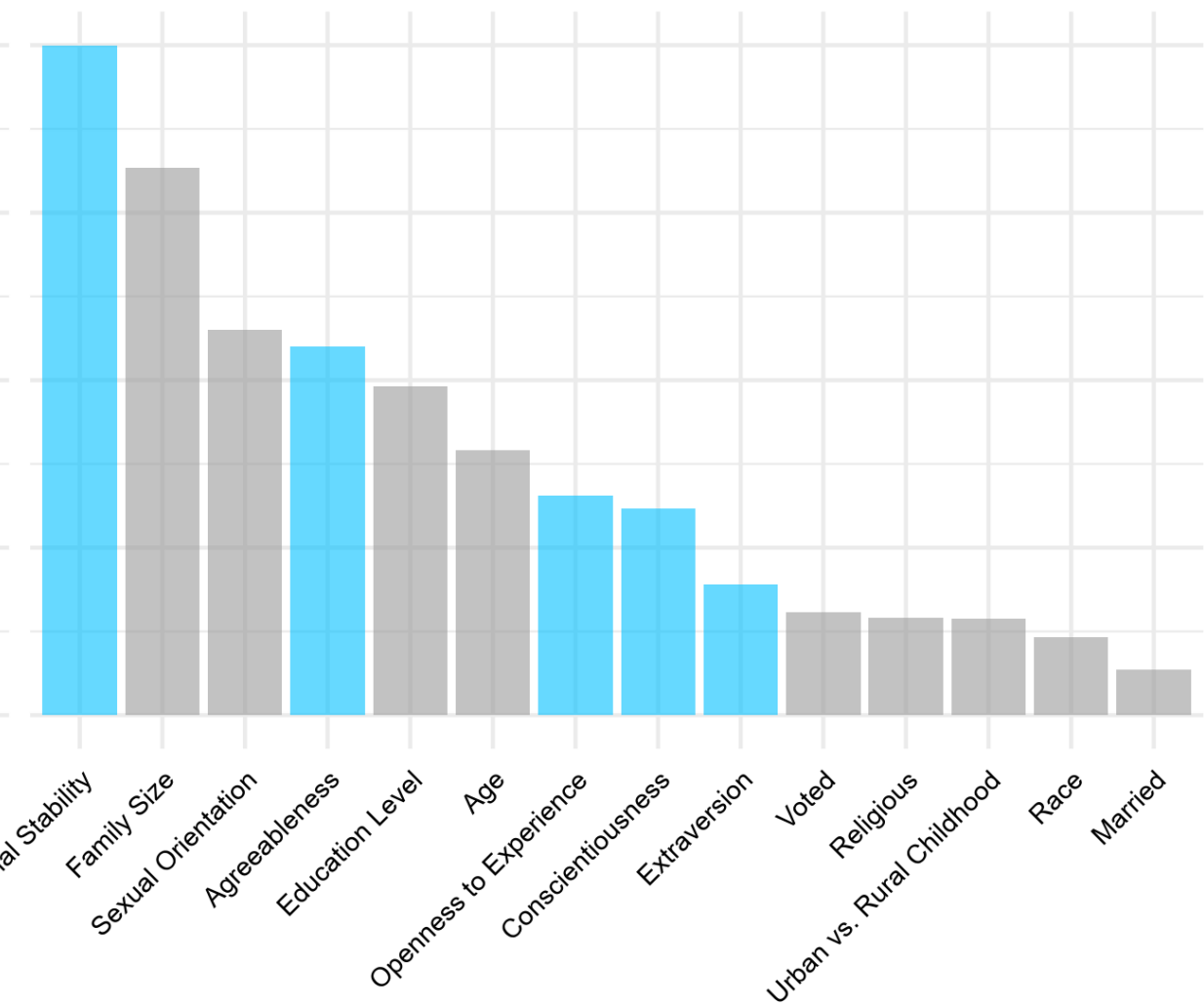


Demographic

Ordered Descending by Female Rank



Ordered Descending by Male Rank



Category ■ Personality, Female ■ Personality, Male ■ Demographics / Other

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If important features vary by gender, what might this suggest about learning a clinically aligned vocabulary?

- For males, a broad emotional vocabulary to parse a variety of emotions
- For females, more important to focus on stress and anxiety to parse depression

Takeaways

DASS-21

- Important depression features meaningfully different for males and females
 - Interrelated features suggests anxiety and stress have an amplifying effect on depression for females, but not males

Demographic

- Personality features generally more important for females
 - Suggests pathway to emotional stability is through multiple personality traits for females.
 - Potentially one reason why stress and anxiety are interrelated with depression.
- Emotional stability generally more important for males
 - Suggests pathway to stability is through managing emotional volatility.



Anything that's human is mentionable, and anything that is mentionable can be more manageable. When we can talk about our feelings, they become less overwhelming, less upsetting, and less scary.

~ Fred Rogers

Thank you!

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